Introduction

Dirichlet regression models compositional outcomes — vectors of proportions that sum to 1, such as relative cell-type composition in a tissue, time budgets across activities, microbial relative abundances, or budget shares across categories. The Dirichlet distribution is the multivariate generalisation of the Beta and lives on the unit simplex; Dirichlet regression places covariate effects on each component while preserving the unit-sum constraint inherent in compositional data.

Prerequisites

A working understanding of beta regression, the unit simplex, and compositional data as fundamentally different from independent proportions.

Theory

A composition $Y = (Y_1, \ldots, Y_k)$ with $\sum_j Y_j = 1$ and $Y_j > 0$ follows a Dirichlet distribution $\mathrm{Dir}(\alpha_1, \ldots, \alpha_k)$. Dirichlet regression parameterises the component means as

$$\mu_j = \frac{\exp(x^\top \beta_j)}{\sum_l \exp(x^\top \beta_l)},$$

with $\alpha_j = \mu_j \phi$ where ϕ is a precision parameter. The structure mirrors multinomial logistic regression, with one reference component fixed for identifiability and the other $k-1$ sets of coefficients estimated.

For very small samples or high precision, the Dirichlet can be unstable; weakly informative Bayesian priors (`brms` with `family = dirichlet()`) are an alternative.

Assumptions

Components sum to 1 with all components strictly positive (zeros require perturbation or zero-inflated extensions); independent observations; the Dirichlet variance structure is appropriate for the data.

R Implementation

```r
library(DirichletReg)

set.seed(2026)
n <- 150
x <- rnorm(n)
a1 <- exp(0.3 * x); a2 <- exp(-0.2 * x); a3 <- exp(0.5 * x)
Y <- t(apply(cbind(a1, a2, a3), 1,
             function(a) {
               phi <- 20
               rd <- rgamma(length(a), a * phi, 1)
               rd / sum(rd)
             }))

d <- data.frame(Y = DR_data(as.data.frame(Y)), x = x)
fit <- DirichReg(Y ~ x, data = d)
summary(fit)
```

Output & Results

`DirichReg()` returns coefficients per non-reference component on the multinomial logit scale plus the precision parameter. Predictions back on the simplex come from `predict()`; they always satisfy the unit-sum constraint by construction.

Interpretation

A reporting sentence: "Dirichlet regression showed that x positively associated with components 1 and 3 (multinomial-logit slopes 0.5 and 0.7 vs the reference) and negatively with component 2 (-0.4); predicted compositions at $x = -1, 0, +1$ shifted from (0.30, 0.40, 0.30) to (0.33, 0.30, 0.37) to (0.36, 0.22, 0.42), preserving the unit-sum constraint." Always communicate compositional results as predicted compositions; raw multinomial-logit coefficients are difficult to interpret.

Practical Tips

- Exact zeros or exact ones in the composition need perturbation before fitting; `compositions::clr()` and `zCompositions::cmultRepl()` provide standard zero-replacement algorithms.
- For compositional data with many components ($k > 6$), consider aggregating to fewer meaningful categories or using additive-log-ratio (ALR) or centred-log-ratio (CLR) transformations followed by ordinary multivariate regression.
- For zero-inflated compositions (many exact zeros), use zero-inflated Dirichlet or Zadeh-Aitchison ZI-logratio models.
- Interpretation is cleanest via predicted compositions plotted against covariate values; raw coefficients on the multinomial-logit scale are rarely communicative.
- For paired or longitudinal compositional data, a paired Dirichlet model or a mixed-effects extension is needed; ordinary Dirichlet regression assumes independence across observations.
- For Bayesian Dirichlet regression with weakly informative priors, `brms(family = dirichlet())` provides a familiar interface.

R Packages Used

`DirichletReg` for the canonical `DirichReg()` implementation; `compositions` and `zCompositions` for compositional data pre-processing; `brms` for Bayesian Dirichlet regression; `aldex2` for differential-abundance compositional analysis in microbiome work.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- Correlation vs regression; Model-I/II regression
- Simple linear regression
- Multiple regression: confounding, interaction, centring
- Diagnostics: residuals, QQ, leverage, Cook's distance, VIF
- Robust regression, weighted LS, HC (sandwich) standard errors
- One-way ANOVA and contrasts (emmeans)
- Two-way / factorial ANOVA with interaction
- RCBD and repeated measures → mixed models
- GAMs with mgcv
- Non-linear regression with nls
- Logistic regression (binomial GLM)

- ANCOVA in RCTs (baseline adjustment)
- Ordinal and multinomial regression
- Poisson and negative-binomial regression
- Calibration, discrimination, ROC/AUC, Brier score
- Dichotomisation, change scores, regression to the mean
- Decision curves, NRI, IDI